Hypothesis Testing

https://en.wikipedia.org/wiki/Statistical_hypothesis_testing

Purpose of hypothesis testing is to determine whether there is enough statistical evidence in favor of a certain belief/hypothesis about a parameter value or a distribution.

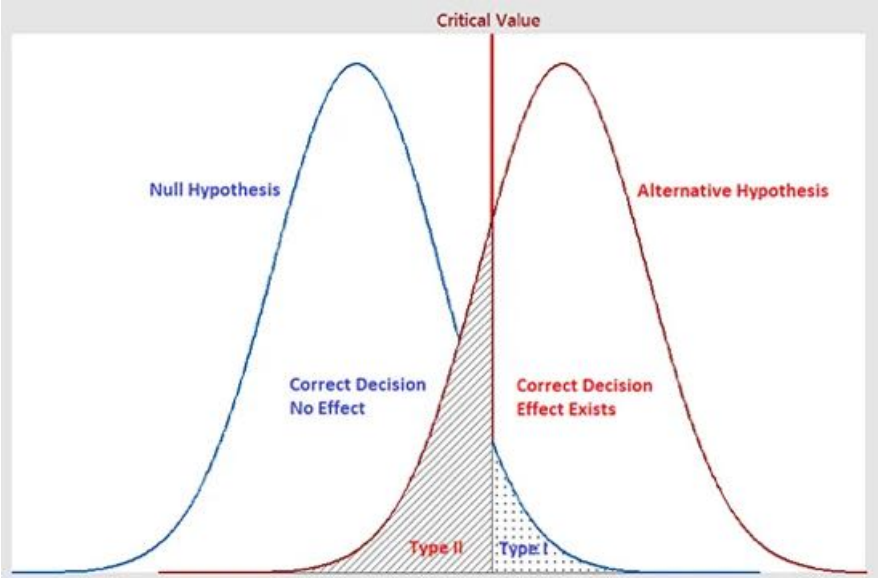
Four steps:

Step1:
Formulate hypotheses – usually at least two - the main one (usually called "Null Hypothesis") and an alternative.

Step2:
Decide which level of significance we need. It is usually denoted as **alpha** or **α**, and means the probability of rejecting the null hypothesis when it is true. Most common choice for the value of alpha is 0.05.

Step3:
Select the test type and compute the test statistic. There are many different types of tests in statistics like t-test,Z-test,chi-square test, anova test ,binomial test, one sample median test, etc.

Step 4:
Make a decision.



Type I error - false rejection of the null hypothesis
Type II error - false acceptance of the null hypothesis

p-value & null hypothesis

A **p-value** (probability value) is a probability of getting results at least as extreme as yours. If this probability is very small, this means that the result is unlikely to be explained by the null hypothesis. So we should reject the null hypothesis.

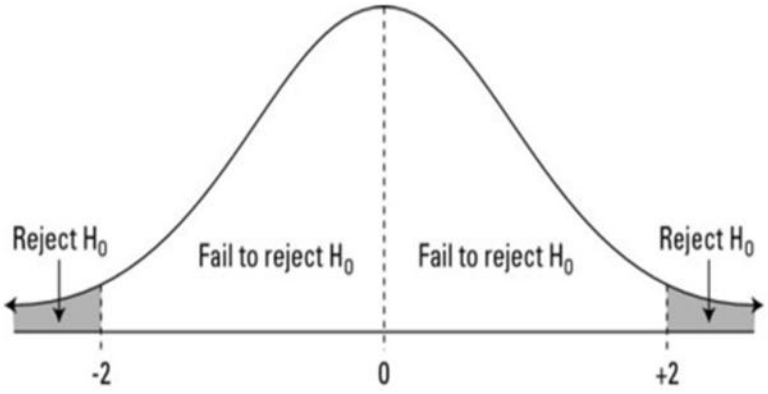
```
P-value    probability of observed
-------------------------------
P < 0.05          5%
P < 0.01          1%
p < 0.001         0.1%
-------------------------------
```

A small p-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, so you reject the null hypothesis. People say that this evidence is "statistically significant".

Level of significance (**alpha** or **α**, typically 0.05) - probability that the event could have occurred by chance.

Confidence Interval (CI). A 95% CI is the interval that you are 95% certain contains the true population value as it might be estimated from a much larger study.
- https://en.wikipedia.org/wiki/Confidence_interval

z-score is the distance from the mean measured in the number of standard deviations.



Type I error - false rejection of the null hypothesis
Type II error - false acceptance of the null hypothesis

\ Decision Truth \	Decision to Accept Ho	Decision to Reject Ho
Ho is True	Correct Decision	type I error
Ho is False	type II error	Correct Decision

Checking whether a coin is fair

https://en.wikipedia.org/wiki/Checking_whether_a_coin_is_fair

Null Hypothesis (H0) : the coin is fair (flips heads/tails evenly).
Alternative (H1): the coin flips heads/tails unevenly with error **E = |P_h-0.5|**, where P_h is the probability of heads.

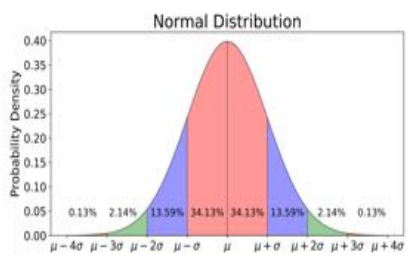
Problem 1

We tossed a coin 10 times and got 8 heads. Compute p-value.

We just need compute the probability of 8 or more heads in 10 tosses (assuming the coin is fair).
Note that the number of heads follows a binomial distribution with n=10 and p=0.5.
So the p-value = P[8] + P[9] + P[10] = .044 + .01 + .001 = 0.055.

Coin flipping leads to binomial distribution,
which can be approximated with normal Gaussian distribution.
Here are confidence levels for Z (standard normal distribution):

Z value	Confidence level	Comment
0.6745	50.000%	Half
1.0000	68.269%	One std dev
1.6449	90.000%	"One nine"
1.9599	95.000%	95 percent
2.0000	95.450%	Two std dev
2.5759	99.000%	"Two nines"
3.0000	99.730%	Three std dev
3.2905	99.900%	"Three nines
3.8906	99.990%	"Four nines"
4.0000	99.993%	Four std dev
4.4172	99.999%	"Five nines"



Problem 2

How many times should we flip a coin to test the null hypothesis.

Suppose we flipped "n" times and got "x" heads.
The estimate of the probability of heads P_h = x/n.
Suppose "x" follows binomial distribution with probability "p" of heads.
At large "n" is is approximated with normal distribution.

$$f(x) = C(n,x)p^x q^{n-x} \sim \frac{1}{\sqrt{2\pi npq}} e^{-(x-np)^2/2npq}$$

We use character "**σ**" to denote the standard error of estimating x:

σ^2 = npq

Note that it has max value when p = q = 0.5, so **σ^2= n/4**
So we can estimate the error for 2**σ** (~ 95% confidence) as:

2σ ~ √n

This is very easy to remember – error grows as square root of n.
If we estimate error for ratio "x/n", it will be then: $\frac{1}{2\sqrt{n}}$

And hence the value of maximum error (E) is given by:

$$E = Z\, s_p = \frac{Z}{2\sqrt{n}}$$

Or:

$$n = \frac{Z^2}{4\,E^2}$$

For example, if we want confidence 95% (Z=1.9599), then

- for E=0.5 we need n = (1.9599)**2 / (4*0.5**2) = 4 flips
- for E=0.25 we need n = (1.9599)**2 / (4*0.25**2) = 16 flips
- for E=0.01, we need n = (1.9599)**2 /(4*0.01**2) = 9,603 flips

Binomial Test in Python

```
# binomial distribution
```

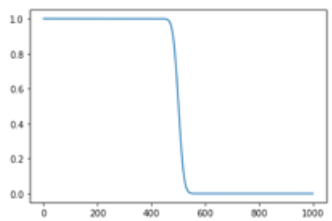
pmf = **probability mass function** - gives the probability that a discrete random variable is exactly equal to some value

Binomial PMF : $$f(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

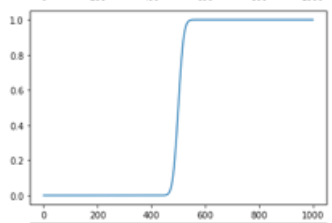
```
from scipy.stats import binom

binom.pmf(0,2,0.5) # 0.25
binom.pmf(1,2,0.5) # 0.5
binom.pmf(2,2,0.5) # 0.25

n=10
arr = []
for ii in range(n+1):
    arr.append(binom.pmf(ii, n, 0.5))
print(sum(arr))  # 1
print(arr)

# 8, 9, 10 heads out of 10 trials:
binom.pmf(8,10,0.5)  # 0.043945
binom.pmf(9,10,0.5)  # 0.009766
binom.pmf(10,10,0.5) # 0.000977
                # sum: 0.054688
```

```
# We flip a coin N=1000 times.
# If x is number of heads, it has binomial distribution with sigma = sqrt(N)/2 = 15.81
# If it lands on heads 531 times, it is within 2*sigma.
# But let's perform a binomial test for null hypothesis that coin is fair:
binom_test(x=531, n=1000, p=1/2, alternative='greater') # P = 0.0268
# Because this p-value (0.0268) is less than 0.05, we reject the null hypothesis.
```

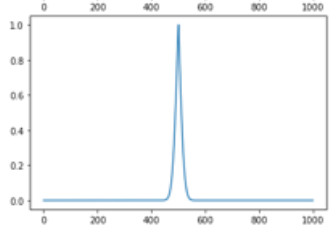
```
from scipy.stats import binom_test  # returns p-value of test
```



```
binom_test(x=0,   n=1000, p=1/2, alternative='less')  # 1
binom_test(x=470, n=1000, p=1/2, alternative='less')  # 0.973
binom_test(x=500, n=1000, p=1/2, alternative='less')  # 0.5
binom_test(x=530, n=1000, p=1/2, alternative='less')  # 0.031
binom_test(x=999, n=1000, p=1/2, alternative='less')  # 0
```



```
binom_test(x=0,   n=1000, p=1/2, alternative='less')  # 0
binom_test(x=470, n=1000, p=1/2, alternative='less')  # 0.031
binom_test(x=500, n=1000, p=1/2, alternative='less')  # 0.5
binom_test(x=530, n=1000, p=1/2, alternative='less')  # 0.973
binom_test(x=999, n=1000, p=1/2, alternative='less')  # 1
```

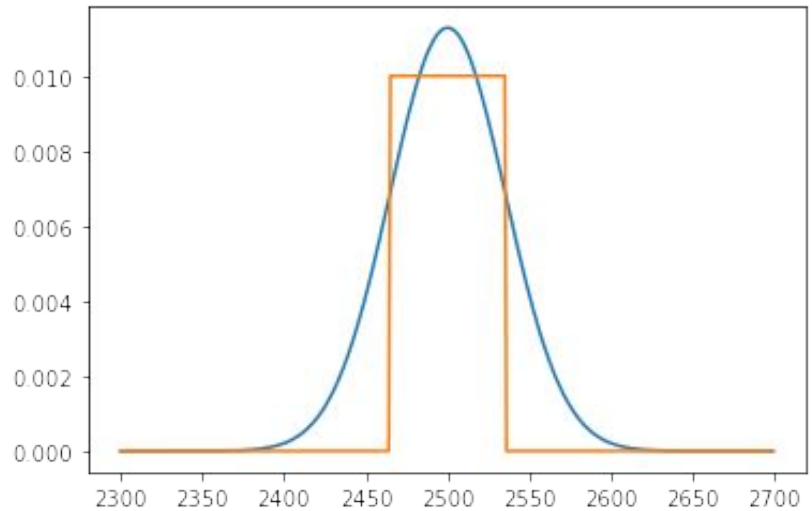


```
binom_test(x=0,   n=1000, p=1/2, alternative='two-sided')  # 0
binom_test(x=470, n=1000, p=1/2, alternative='two-sided')  # 0.062
binom_test(x=500, n=1000, p=1/2, alternative='two-sided')  # 1.0
binom_test(x=530, n=1000, p=1/2, alternative='two-sided')  # 0.062
binom_test(x=999, n=1000, p=1/2, alternative='two-sided')  # 0
```

```
# We flip a coin 10 times and it lands on heads 8 times.
# Perform a binomial test to determine if the coin is biased.
binom_test(x=8, n=10, p=1/2, alternative='greater') # P = 0.0546875
# Because this p-value (0.0546875) is not less than 0.05,
# we accept the null hypothesis.
```

Binomial Distribution – Plot, Sigma

```
import os, sys, math
from scipy.stats import binom_test
import matplotlib.pyplot as plt
xx  = []
yy  = []
yy2 = []

sigma = math.sqrt(5000)/2.0
v1 = 2500-sigma
v2 = 2500+sigma
print(v1,v2)    # 2464.64, 2535.36

for ii in range (2300, 2700):
    xx.append(ii)
    yy.append(binom.pmf(ii,5000,0.5))
    val = 0.0
    if v1 <= ii <= v2:
        val = 0.01
    yy2.append(val)
_ = plt.plot(xx,yy)
_ = plt.plot(xx,yy2) # show sigma range
```



Checking whether a coin is fair - χ^2 test

Problem:
We want to test a null hypothesis that the coin is fair.
We flip the coin N times and got H heads.
So number of observed tails T = N-H.

Expected values are N/2 for heads and for tails.

Using χ^2 goodness of fit test

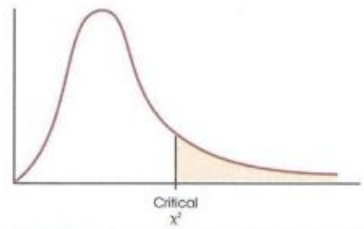
X^2 = sum((observed – expected)^2/expected) = [(H– N/2)^2 / (N/2)] + [(T – N/2)^2 / (N/2)]

The P-value is the probability that a chi-square statistic having one degree of freedom is more extreme than that value.

Specific example:
N=100, H=60, T=40

X^2 = [(60– 50)^2 / 50] + [(40 – 50)^2 / 50] = 10^2 / 50 + 10^2 / 50 = 2+2 = 4

X^2 distribution for 1 degree of freedom gives 3.84 for P-value P=0.05.
We have 4, which is more extreme.
So we reject the null hypothesis (that the coin is fair).



χ^2

	Proportion in Critical Region				
df	0.10	0.05	0.025	0.01	0.005
1	2.71	3.84	5.02	6.63	7.88
2	4.61	5.99	7.38	9.21	10.60
3	6.25	7.81	9.35	11.34	12.84
4	7.78	9.49	11.14	13.28	14.86
5	9.24	11.07	12.83	15.09	16.75
6	10.64	12.59	14.45	16.81	18.55
7	12.02	14.07	16.01	18.48	20.28
8	13.36	15.51	17.53	20.09	21.96
9	14.68	16.92	19.02	21.67	23.59
10	15.99	18.31	20.48	23.21	25.19

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Checking whether a coin is fair - t-test

I am not sure if the t-test can be used for coin flipping hypothesis testing.

t-test is used for data which is presumed to have normal distribution, and only for small samples of data (< 30 points).

But here is formal application of t-test:

Problem:
We want to test a null hypothesis that the coin is fair.
We flip the coin N times and got H heads.
So number of observed tails T = N-H.
Expected numbers are N/2 for heads and for tails.

To calculate t-test we assume values 1 for heads and -1 for tails.
Then suppose N=100, H=60, T=40

Sample Mean X = (H – T)/100 = 20/100 = 0.2
Expected mean = N/2 – N/2 = 0
Sample Variance:
$S^2 = (1/99) * (60*(1-0.2)^2 + 40*(-1-0.2)^2) = 0.969$
Sample Standard Deviation:
S = sqrt(0.9696) = 0.9847
t-statistics:
(0.2– 0) / (0.9847 / 10) = 2.031

Student t-test

Let X_i be independently and identically drawn from the normal distribution with expected mean value **μ** and variance **σ^2**

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$ ----- Sample mean

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$ ----- Sample variance

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$ Random variable, Normal distribution

$$\frac{\bar{X} - \mu}{S/\sqrt{n}},$$ Random variable, **t-distribution**.
It uses S = sqrt(sample variance) because real sigma is unknown. This creates **uncertainty** – and leads to the "**fat tails**" in the t-distribution.

	Proportion in One Tail				
	0.25	0.10	0.05	0.025	0.01
	Proportion in Two Tails Combined				
df	0.50	0.20	0.10	0.05	0.02
1	1.000	3.078	6.314	12.706	31.821
2	0.816	1.886	2.920	4.303	6.965
3	0.765	1.638	2.353	3.182	4.541
4	0.741	1.533	2.132	2.776	3.747
5	0.727	1.476	2.015	2.571	3.365

Hypothesis Testing Examples

Problem1:

Given: women's heights are normally distributed with the mean 65” and standard deviation of 3.5”
Question: What is the probability of finding a random sample of 50 women with a mean height of 70”.

To answer, let's estimate the **z-score**:
$z = (x - \mu) / (\sigma / \sqrt{n}) = (70 - 65) / (3.5/\sqrt{50}) = 5 / 0.495 = 10.1$

We know that 99% of values fall within 3σ (68, 95, 99.7 rule).
We have z-score = 10, which is much more than 3.
So the probability that any sample of 50 women will have a mean height of 70” will be much-much less than 1%.

Problem 2:

Suppose you are testing a claim that the percentage of all women with varicose veins is 25%, and your sample of 100 women had 20% with varicose veins.
Then the sample proportion p=0.20.
The standard error for your sample percentage is the square root of p(1-p)/n which equals 0.04 or 4%.

You find the test statistic by taking the proportion in the sample with varicose veins, 0.20, subtracting the claimed proportion of all women with varicose veins, 0.25, and then dividing the result by the standard error, 0.04.

These calculations give you a test statistic (standard score) of –0.05 divided by 0.04 = –1.25.

This tells you that your sample results and the population claim in H0 are 1.25 standard errors apart; in particular, your sample results are 1.25 standard errors below the claim.

When testing H0: $p = 0.25$ versus Ha: $p < 0.25$, you find that the p-value of -1.25 by finding the probability that Z is less than -1.25. When you look this number up on the **Z-table**, you find a probability of 0.1056 of Z being less than this value.

Note: A **Z-table** is simply a table of values of the cumulative distribution function of the normal distribution.

The dichotomous use of p-value in statistical inference is worse than useless.

1. p-value is a conditional probability, i.e., **Pr(Data | H0 is true)**
   but what we want to know is **Pr(H0 is true | Data)**
2. the calculation of p-value depends on at least three things:
   - the effect size (e.g., difference of two means due to two treatment, odds ratio, etc.),
   - the uncertainty level of the point estimate (e.g., standard deviation),
   - and sample size
3. it is a standard result in statistics theory that, based on one set of sample data we cannot tell the observed effect **size is due to treatment or due to random variation** (i.e., sampling distribution requires we repeat the same experiment many times before any confirmatory conclusion can be made)
4. if by nature the H0 is false (in the majority cases of real life data / research questions), p-value =0.

In summary, P-value based Null Hypothesis Significance Test (NHST) is logically not defensible, technically flawed and practically damaging

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https://www.researchgate.net/post/What_is_the_relationship_between_R-squared_and_p-value_in_a_regression